Incantation: Natural Language is the Action Interface for Multi-Entity Video World Models

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Abstract

Controllable video world models have converged on an interface—keyboard, mouse, or discrete action IDs—that structurally cannot represent what multi-entity interactive video requires: independent per-entity channels, semantic composition across entities, and a shared representation across heterogeneous worlds. We argue that **natural language is the action interface for multi-entity video world models**. Four structural properties—per-entity independence, semantic compositionality, world-agnostic universality, and pretrained semantic priors—predict two behaviours that no discrete-ID interface can replicate: *graded cross-entity semantic transfer*, and *open-vocabulary compositional prompt coverage*. A controlled head-to-head between a language-conditioned and an otherwise-identical discrete-ID variant confirms both: the two variants tie on in-distribution control, diverge by more than an order of magnitude on cross-entity transfer ($\sim$80% vs. $\sim$6% uniform baseline), and the ID variant is structurally undefined on out-of-vocabulary prompts. We present **Incantation**, which realises this paradigm with two general architectural patterns—bidirectional history with decoupled text cross-attention for per-frame language conditioning, and Self-Forcing distillation with RoPE-decoupled KV-cache sliding for real-time long-horizon inference—and validate them on a three-world testbed (two *Elden Ring* bosses with disjoint move sets, and *The King of Fighters*) chosen for the frame-accurate multi-entity action labels games uniquely provide. The system achieves $\sim$90% action control accuracy with zero architecture modification across all three worlds and sustains 480p at 16 fps on a single H100 for 10+ minutes of continuous real-time generation.

Figure 1: Demo

1 Introduction

Video diffusion models [21, 6, 39] have raised the visual quality ceiling of generated video to near-cinematic fidelity, and a growing body of work now asks them to act as *controllable interactive world models* [7, 26, 27, 3, 13, 47, 20, 40, 2]. The visible progress on quality has obscured a quieter but more fundamental question this progress has left unanswered: *what is the right interface through which such a model should be controlled?*

The interface question. Every existing controllable video world model inherits one of two choices. **Device-level inputs** (keyboard, mouse, gamepad)—used by Genie [7, 26, 27], Oasis [2], Matrix-Game [20], and GameNGen [40]—are tied to a single human operator and inherit the input ontology of a specific engine. **Discrete action IDs**—used by DIAMOND [3] and most earlier game-specific generators—encode each action as an opaque integer embedding with no shared structure across actions, entities, or worlds. Both choices collapse under even modest generalizations of the setting: a second autonomous entity has no keyboard of its own; two generators trained on different worlds cannot share their embedding tables; an action that one entity has never performed in training has no input slot through which to invoke it. The device-level and ID paradigms are not merely underpowered; they are *structurally unable* to represent the behaviours that a multi-entity, long-horizon interactive video world model is supposed to support.

Any conditioning interface that hopes to survive these generalizations must simultaneously provide **(i)** per-entity independence (each entity is controlled through its own channel), **(ii)** semantic compositionality (action representations carry meaning that transfers across entities), **(iii)** world-agnostic universality (the same interface works across titles and engines without re-engineering the input layer), and **(iv)** access to pretrained semantic priors (action names are already partially meaningful before task-specific training). No device-level or discrete-ID scheme satisfies any of these four by construction.

Natural language is the action interface. We argue that *natural language is the only representation that satisfies all four requirements*, and that this choice has consequences reviewers of this paper should be able to predict and verify empirically. Structured per-frame language prompts route through a pretrained text encoder whose subword embeddings are already meaningful before a single gradient step on gameplay data; syntactic slots provide independent per-entity channels; composition of tokens (`‘‘tail’’ + ‘‘of the’’ + entity-name`) provides an action space with an interpolable geometry; and the same template extends verbatim across worlds whose action vocabularies share not a single ID. These four structural properties jointly predict two falsifiable behaviours that no discrete-ID interface can exhibit by construction:

1. **Graded cross-entity semantic transfer.** Prompting entity E with an action learned only for E' should produce semantically coherent output grounded in E's visual repertoire, rather than random behaviour or collapse to E's training modes. Discrete-ID embeddings are orthogonal by construction and offer no gradient along which such transfer could occur.
2. **Open-vocabulary compositional prompt coverage.** Prompts that extend, paraphrase, or compose the training vocabulary should remain parseable and controllable. Discrete-ID interfaces are not merely worse on such prompts—they are *structurally undefined*, since no index in a fixed-dimensional embedding table corresponds to a prompt the vocabulary did not enumerate.

A controlled head-to-head (Section 4.2.2) matches a language-conditioned variant against an otherwise-identical discrete-ID variant sharing a joint action vocabulary, identical architecture, data, and compute. The two variants tie on in-distribution control accuracy, ruling out capacity or optimization as explanations, and diverge by more than an order of magnitude on cross-entity transfer ($\sim 80\%$ vs. $\sim 6\%$ uniform baseline). The ID variant is structurally undefined on out-of-vocabulary prompts. Together, these observations establish that language's advantage is not a quality-of-conditioning gap but a *representational-structure gap* that no amount of ID training data or compute can close.

Contributions. We validate the paradigm on a three-world testbed—two *Elden Ring* bosses with disjoint move sets (3D action RPG, photorealistic rendering) and *The King of Fighters* (2D fighting,

pixel-art, side-scrolling camera)—chosen because frame-accurate multi-entity action labels at interactive frame rates are what games uniquely provide and what no current non-game interactive-video domain offers. The paradigm itself is world-agnostic; extending the testbed to non-game interactive video is future work. Concretely, we contribute:

1. **A paradigm claim with falsifiable predictions.** We identify four structural properties under which natural language is the only representation satisfying the requirements of multi-entity video world models, derive the two predictions above, and verify both in a controlled NL-vs-ID head-to-head under a joint action vocabulary, identical architecture, and identical compute: $\sim$80% cross-entity transfer (uniform baseline $\sim$6%) and open-vocabulary coverage where IDs are structurally undefined (Sections 4.2.1, 4.2.2).
2. **Two architectural patterns for language-conditioned multi-entity video world models.** *Per-frame language conditioning on a bidirectional backbone* via bidirectional history self-attention with text cross-attention decoupled to the noisy target (Section 3.1), and *real-time long-horizon streaming inference* via two-stage distillation (ODE initialisation + Self-Forcing) with RoPE-decoupled KV-cache sliding (Section 3.2). Both are formulated independently of world or genre.
3. **End-to-end validation on a three-world testbed.** A single unmodified conditioning architecture runs across all three worlds with $\sim$90% action control accuracy, sustains 480p at 16 fps on a single H100 for 10+ minutes of continuous real-time generation, and exhibits the cross-world universality that device-level and discrete-ID interfaces structurally cannot provide (Section 4.2.3).

Open release. To facilitate further work on the conditioning-interface question, **we will release all code, trained checkpoints, and the full frame-accurate multi-entity action dataset** (80+ hours across three worlds, with per-frame per-entity action labels extracted from engine memory).

2 Related Work

2.1 Controllable Video World Models

Early world models compressed game observations into compact latent spaces for planning [17, 18]. The shift to diffusion-based rendering began with GameNGen [40], the first real-time diffusion game engine (DOOM, 20 fps), and DIAMOND [3], which extended this to Atari. These systems are limited to single-entity discrete conditioning with no game-logic enforcement. GameGen-X [9] and GameFactory [46] generalize diffusion-based game generation to open-domain scenes with keyboard/mouse action control, yet both remain single-player systems that operate with low-level input signals rather than semantic entity-level commands.

The Matrix [13] and the Matrix-Game series [47, 20] scale to AAA-quality video at 16–25 fps, but target open-world *navigation* (ego-motion only) or Minecraft (WASD + mouse), both of which lack opponent modeling and episode-level game state. The Genie series [7, 26, 27] learns latent action representations from unlabeled video and achieves multi-minute 3D world generation, but accepts only raw keyboard/mouse inputs and provides no interface for independently commanding a second entity with a semantic vocabulary—they are *world generators*, not game engines. WorldPlay [35] and Infinite-World [43] push streaming video world models to 720p/24 fps and 1000-frame horizons with long-term geometric consistency, but still accept only raw keyboard/mouse inputs with no multi-entity interface. Vid2World [22] and Hunyuan-GameCraft-2 [36] extend interactive world modeling to action-conditioned and instruction-following generation respectively, but both remain single-agent systems without independent opponent control. AVID [29] adapts closed-source video diffusion models via learned masks without parameter access, and MineWorld [16] replaces diffusion with a visual-action autoregressive Transformer for real-time Minecraft at 4–7 fps; both remain single-agent with no opponent modelling. Oasis [2] demonstrates real-time Minecraft generation. Solaris [4] extends this to two human players but generates *independent* per-player streams from separate first-person viewpoints, so mutual adversarial interaction within a shared frame is never jointly modeled. COMBAT [1] trains a 1.2 B DiT world model on Tekken 3 and learns reactive opponent behaviour emergently from single-player data, but provides no explicit interface for independently commanding the opponent's strategy. MultiGen [28] introduces external memory for editable diffusion game engines and symmetric real-time multiplayer rollouts, but only provides limited first-person views rather than full scenes and requires multiple models to control multiple subjects. ShareVerse [49] achieves multi-agent consistent video generation for shared-world autonomous

Table 1: Systematic comparison of interactive video world models. ✓ = supported, ✗ = not supported, $\sim$ = partial. *Multi-entity* requires independent control of two entities with distinct action vocabularies. *Semantic NL* requires per-frame natural language action descriptions.

Model	Multi-entity	Semantic NL	Logic Rules	Opponent	$\geq$16fps	$>$5min
GameNGen [40]	✗	✗	✗	✗	✓	✓
Genie 2 [26]	$\sim$	✗	✗	✗	✓	✗
Genie 3 [27]	$\sim$	✗	✗	✗	✓	$\sim$
The Matrix [13]	✗	✗	✗	✗	✓	✓
Matrix-Game 2.0 [20]	✗	✗	✗	✗	✓	$\sim$
LingBot-World [14]	✗	✗	✗	✗	✓	✓
Solaris [4]	$\sim$	✗	✗	✗	$\sim$	$\sim$
Incantation (Ours)	✓	✓	✓	✓	✓	✓

driving, while LiveWorld [12] models persistent entity evolution when unobserved; neither addresses interactive adversarial game combat with semantic control. LingBot-World [14] achieves 16 fps long-horizon stability on navigation scenarios but provides only WASD/text control without opponent modeling.

Against this body of work, **Incantation** makes a distinct claim: the primary bottleneck in scaling controllable video world models to multi-entity, long-horizon, cross-world settings is not the backbone but the *conditioning interface*. Every system above inherits either device-level inputs (tied to a single operator and a specific engine's ontology) or discrete action IDs (orthogonal embeddings with no semantic structure across actions, entities, or worlds). We argue—and empirically demonstrate—that natural language is the only representation that simultaneously supports independent per-entity channels, semantic composition across entities, world-agnostic universality, and access to pretrained semantic priors. To our knowledge no prior video world model provides independent per-frame semantic control of two heterogeneous entities within a shared adversarial viewpoint; Table 1 summarises the comparison.

2.2 Efficient Streaming Video Generation

Self-Forcing [10] eliminates exposure bias for autoregressive video generation. StreamingLLM [45] and LM-Infinite [19] introduce attention-sink tokens and sliding-window KV caches to bound memory in language models while keeping position indices within a trained range. Flow matching [25] and consistency distillation [34] reduce diffusion inference steps. Our RoPE-decoupled KV-cache sliding adapts the attention-sink principle to autoregressive video diffusion under two constraints absent from language settings: keys must be stored *pre-rotation* so that local positions can be reassigned on-the-fly without cache invalidation, and the sliding window size is calibrated to the typical duration of an entity-level action rather than to linguistic context length. Together, these adaptations yield the first streaming method to jointly address memory boundedness, positional OOD prevention, and real-time throughput in an interactive video world model.

3 Method

Realising the language-as-action-interface paradigm at end-to-end scale requires solving two architectural problems that any language-conditioned multi-entity video world model encounters, independent of world or genre. We contribute one general pattern for each (Figure 2). **Stage 1** (Section 3.1): *per-frame language conditioning on a bidirectional backbone*, via a per-entity prompt formulation, bidirectional history self-attention, and text cross-attention decoupled to the noisy target only. **Stage 2** (Section 3.2): *real-time long-horizon streaming inference*, via two-stage distillation (ODE initialisation + Self-Forcing) and RoPE-decoupled KV-cache sliding.

3.1 Stage 1: Language-Conditioned Architecture

As established in Section 1, device-level inputs cannot independently condition multiple heterogeneous entities, and discrete action IDs lack the semantic structure needed for cross-entity generalization. We therefore adopt natural language as the sole action interface. This subsection presents

Figure 2: The two architectural patterns instantiated in **Incantation**. **Left**: per-frame language conditioning on a bidirectional backbone via decoupled text cross-attention (Stage 1). **Right**: real-time streaming inference via Self-Forcing distillation and RoPE-decoupled KV-cache sliding (Stage 2). Both patterns are formulated independently of the world being modelled.

three coupled design decisions: how prompts are formulated, how attention is structured to convert high-level prompts into frame-accurate actions, and how positional encodings are assigned to enable both in-distribution training and bounded inference.

Prompt Formulation

Each 0.25 s action window is described by a structured natural-language prompt with two parallel, syntactically isolated slots:

```
Player performs [ACTION_P]. Boss performs [ACTION_B].
```

This template *structurally enforces* both **simultaneous control** and **entity decoupling**: the temporal alignment of the two slots ensures that the model jointly reasons about inter-entity dynamics within each frame, while the syntactic separation preserves independent control pathways for each entity. The template naturally extends to settings with more or fewer entities by appending or omitting slots, with no architectural modification required—a flexibility that no fixed-dimensional discrete-ID interface can provide.

Excluding passive-response events from the prompt vocabulary. Passive physiological responses—in our setting, taking-hit events—are not deliberate actions the controller issues but involuntary responses to events happening to an entity. Including them in the conditioning space introduces involuntary pauses that corrupt the model's understanding of volitional movement. We therefore exclude all such passive-response labels from the prompt vocabulary: the prompt vocabulary describes *what an entity chooses to do*, not what happens to it.

Context Assembly

We adopt a *Sink* + *Recent* + *Noisy* context structure. For each training step targeting frame t:

- **Sink frame** (K_s=1): the first frame of the episode, anchoring global context (arena geometry, character appearance) following the attention-sink mechanism of Xiao et al. [44].
- **Recent frames** (K_r=7): the 7 most recent clean latent tokens preceding t. Each latent token corresponds to 0.25 s of gameplay after the base model's VAE temporal compression, so the recent context spans 1.75 s of game time. We ablate K_r in Section 4.3.1.
- **Noisy target** (K_n=1): the partially-denoised latent of frame t.

Attention Design

The conventional approach—causal self-attention over all visual tokens plus full text cross-attention—introduces two failure modes under per-frame language conditioning: **(1) Destruction of pre-trained priors.** The Wan 2.2 base model was pre-trained with full bidirectional attention; its

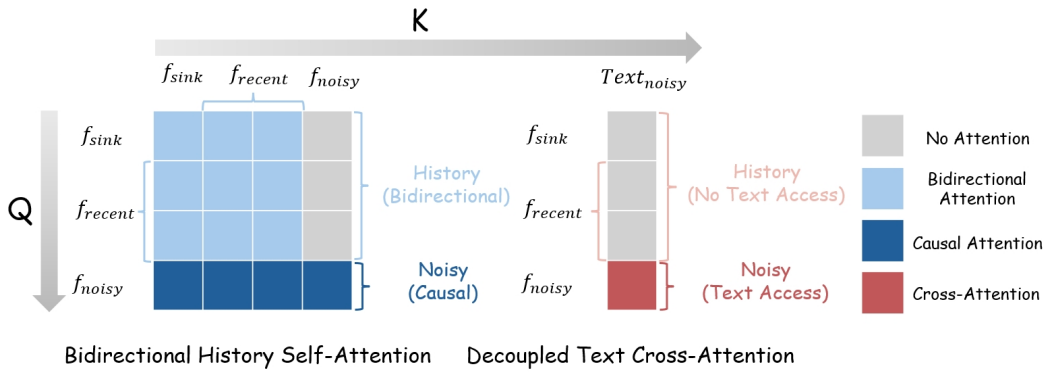

Figure 3: **Attention design.** Bidirectional self-attention is retained over history frames to preserve the spatio-temporal priors of the pre-trained base model. Action cross-attention is restricted exclusively to the noisy target frame, preventing temporal cross-contamination. Together, these two constraints improve per-frame controllability without degrading generation quality.

weights encode symmetric co-occurrence statistics. Imposing a global causal mask discards these priors, requiring costly re-adaptation. **(2) Temporal cross-contamination.** Each action prompt a_t describes exclusively what occurs at time t. Allowing a_t to cross-attend to history frames causes it to retroactively corrupt committed past representations, producing spurious action echoes in adjacent frames.

Bidirectional History Attention. We apply **full bidirectional self-attention** over the $(K_s + K_r)$ history tokens, preserving the base model's pre-trained co-occurrence statistics. A causal boundary separates history from the noisy target, enforcing correct temporal ordering at generation time.

Decoupled Text Cross-Attention. The per-frame action prompt a_t cross-attends *exclusively* with the noisy target token; history frames are masked out entirely. This prevents temporal cross-contamination: the current annotation cannot influence committed past representations. Ablations in Section 4.3.1 validate both design choices.

RoPE Position Design

The sink frame is assigned fixed position 0; recent frames receive sliding positions $1, \ldots, K_r$; the noisy target receives $K_r{+}1$. This scheme is *bounded*: every token always carries a position in $\{0, \ldots, 8\}$, the range seen during training. This guarantee serves double duty: it prevents RoPE OOD during training *and* enables the KV-cache sliding mechanism at inference (Section 3.2).

Training

We fine-tune Wan 2.2 TI2V-5B [38] end-to-end on 64×H100 GPUs with FSDP and mixed precision. Training uses a two-resolution curriculum: 256×448 for 1,000 warm-up steps (lr $= 2\times10^{-5}$), then 480×832 for 29,000 steps (lr $= 1\times10^{-5}$), with global batch size 64. The training data is described in Section 4.1.

3.2 Stage 2: Real-Time Streaming Inference

The bidirectional teacher from Stage 1 requires ∼20 denoising steps per frame and attends over a full visual context, making real-time play infeasible. Stage 2 addresses two coupled bottlenecks: reducing per-frame compute via distillation, and bounding per-frame memory via KV-cache sliding while preserving positional coherence.

Streaming Distillation

We distill the teacher into a 2-step causal student in two sub-stages.

ODE Initialization. We initialize a causal student from the teacher's weights and align its predicted velocity field via a flow-matching consistency objective [25]:

$$\mathcal{L}_{\text{ODE}} = \mathbb{E}_{\tau, v_0, \epsilon}\big[\|f_\theta(v_\tau;\ \text{causal}) - f_{\text{teacher}}(v_\tau;\ \text{bidir})\|_2^2\big] \tag{1}$$

This resolves the attention-mask mismatch in 1,000 steps at 480×832 (64×H100, lr$= 5\times10^{-6}$, batch size 128).

Self-Forcing Distillation. Starting from the ODE-initialized student, Self-Forcing [10] collapses inference to **2 steps**. During training the student conditions on its *own* previously generated frames, directly suppressing the compounding errors that would otherwise accumulate during extended play.

RoPE-Decoupled KV-Cache Sliding

Unbounded generation requires bounding VRAM while preserving positional coherence. Naive truncation causes flickering when evicted frames' absolute position indices suddenly disappear. We solve this by storing keys *before* RoPE rotation and applying rotation on-the-fly using local positions:

$$p_i^{\text{local}} = \text{clamp}\big(p_i^{\text{abs}} - \delta,\ 0,\ C\big), \quad \delta = \max\big(0,\ p_{\text{current}}^{\text{abs}} - C\big) \tag{2}$$

$$\text{Attn}(q_t, k_i) = \big(q_t \cdot R(p_t^{\text{local}})\big)\big(k_i^{\text{raw}} \cdot R(p_i^{\text{local}})\big)^\top / \sqrt{d} \tag{3}$$

When the buffer is full, the oldest non-sink frame is evicted; the sink entry is permanently retained. Because the sliding window bounds relative positions to $[0, K_r{+}1]{=}[0, 8]$—exactly the training range established by the RoPE design in Stage 1—long-horizon generation never encounters OOD positions. The local cap $C{=}16$ is a clamping bound; in practice, actual relative positions remain within $[0, 8]$. Together: (a) VRAM is bounded to $\mathcal{O}(K_s + K_r)$; (b) all positions stay in-distribution; (c) evictions produce no visible artifacts.

4 Experiments

4.1 Setup

Why a game testbed. Validating the paradigm claim requires frame-accurate multi-entity action labels at interactive frame rates, and this combination of properties is what games uniquely provide and what no current non-game interactive-video domain offers. Driving and embodied-manipulation datasets lack frame-level entity-wise action annotations; narrative-video datasets lack adversarial multi-entity dynamics. Games, by contrast, expose per-frame animation state through direct engine memory at zero annotation cost, enforce bounded but non-trivial action vocabularies, and provide unambiguous evaluation criteria. The paradigm claim and the two architectural patterns are formulated independently of games; we choose a three-world game testbed—*Elden Ring* (two bosses with disjoint move sets, 3D action RPG, photorealistic rendering) and *The King of Fighters* (2D fighting, pixel-art, side-scrolling camera)—specifically to span rendering pipelines, dimensionalities, camera structures, and action compositions, so that any result produced by a single unmodified architecture across all three is evidence of world-agnosticism rather than of a specific genre.

Memory-Accurate Data Pipeline. In fast-paced adversarial combat, actions unfold at high frequency: even a 0.25 s labeling offset propagates errors through all subsequent annotations, ruling out vision-based annotation. We circumvent this bottleneck via **direct game memory access**: by instrumenting the game process at runtime, we read the engine's internal `current_animation` field for *both* entities on every frame, obtaining ground-truth labels at zero manual cost. For *Elden Ring*, we capture 30 hours of Margit boss-fight gameplay at 30 fps and 720×1280 resolution, downsampled to 16 fps. After filtering non-combat segments and quality-based pruning, we retain **10,000 high-quality 5-second clips**, with the final 10% held out by recording date as the test split. To evaluate cross-boss generalization, we supplement the dataset with [TODO]-hour segments of the Crucible Knight boss; for cross-game evaluation, we additionally collect [TODO]-hour segments from *The King of Fighters*. Raw per-frame annotations are aggregated into 0.25 s windows via majority voting; a one-time manual calibration corrects the minor temporal offset between the video stream and memory reads, yielding precisely aligned action triplets:

$$\{(v_t,\ a_t^{\text{player}},\ a_t^{\text{boss}})\}_{t=1}^{T}, \quad a_t \in \mathcal{A}_{\text{player}} \times \mathcal{A}_{\text{boss}}. \tag{4}$$

$\mathcal{A}_{\text{player}}$ ($|\mathcal{A}_{\text{player}}|$=13) covers idle, directional movement, directional rolls, sweep attack, thrust attack, death, and execution; $\mathcal{A}_{\text{boss}}$ ($|\mathcal{A}_{\text{boss}}|$=30) covers directional movement and 27 distinct combat animations including staff strikes, light-blade attacks, jump maneuvers, charged thrusts, combo sequences, and execution states. Each action pair is verbalized using the prompt template of Section 3.1.

Mirrored naming for the tail-attack minimal pair. Both Margit and the Crucible Knight possess a signature tail attack, but with distinct visual signatures (Margit: an opaque dark tail; Crucible Knight: a luminous energy tail). To enable the single-token minimal-pair protocol used in Sections 4.2.1 and 4.2.2, we adopt a *mirrored* naming convention: Margit's tail swipe is annotated `Tail of the margit` and the Crucible Knight's is annotated `Tail of the crucible`. The two prompts differ by exactly one token—the entity-identity word—while the surrounding syntactic frame ("`Tail of the _`") is held fixed. Crossing one entity's visual context with the other entity's tail-attack token therefore isolates the effect of the entity-identity token on the rendered visual outcome; no other manipulation is required.

Base model. We fine-tune Wan 2.2 TI2V-5B [39] end-to-end as the teacher. The causal student retains the same architecture with causal-masked self-attention.

Metrics.

- **FVD** (lower is better): Fréchet Video Distance computed with I3D features [8].
- **NDTW** (higher is better): Normalized Dynamic Time Warping between CLIP-ViT-H [31] frame embeddings of generated and reference videos, measuring temporal visual fidelity. Formally, $\text{NDTW} = \exp(-d_{\text{DTW}}/(\gamma \cdot |\text{path}|))$ with γ=0.1.
- **ACA** (%): Action Control Accuracy. Human judgment of whether the generated action within each 0.25 s window matches the input prompt; see user study protocol below.
- **LCR** (%): Logical Closure Rate. Fraction of test episodes where the game reaches a terminal state (boss execution or player death) after the requisite hit count.
- **Hit Det. Acc.** (%): Classification accuracy of the hit detection module on a held-out annotated test set.
- **Latency** (ms/frame): Wall-clock inference time per frame on a single H100 80 GB.

User study protocol. All user studies share the same protocol. Five annotators independently watch each generated clip alongside its per-frame prompt sequence and judge, for every 0.25 s window, whether the action visible in the generated video matches the prompt. ACA is reported as the mean percentage of correctly matched windows across annotators and clips.

4.2 Paradigm Validation: Language as the Action Interface

Before turning to architecture-level ablation, we first test the paradigm-level claim that motivates the entire design: that *natural language is the action interface for multi-entity video world models*. The four structural advantages stated in Section 1—per-entity independence, semantic compositionality, world-agnostic universality, and access to pretrained semantic priors—predict three empirically falsifiable behaviours. Section 4.2.1 tests whether language-conditioned generation exhibits *graded cross-entity semantic transfer* on entity–action pairs never observed in training—a behaviour structurally impossible under opaque IDs. Section 4.2.2 runs the controlled head-to-head against a discrete-ID baseline of identical architecture, data, and compute, spanning in-distribution controllability, cross-entity transfer, and out-of-vocabulary prompt coverage. Section 4.2.3 demonstrates that the same unmodified conditioning architecture applies verbatim across structurally dissimilar worlds—3D action RPG combat and 2D pixel-art fighting. Architecture-level ablation for the two patterns (Stage 1 conditioning design, Stage 2 streaming) follows in Sections 4.3–4.4.

4.2.1 Cross-Entity Semantic Transfer

The structural claim of this paper—that language provides semantic compositionality unavailable to discrete-ID interfaces—predicts a specific, falsifiable behavior: a model should be able to interpret action prompts on entities for which the specific entity–action pair was *never* observed in training, by grounding the action in its linguistic semantics rather than entity-specific memorization. We test

Figure 4: **Cross-entity semantic transfer. Top**: Margit executing "Tail of the margit" (in-distribution, a dark opaque tail). **Bottom**: Crucible Knight prompted with "Tail of the margit"—an entity–action pair absent from training; the mirrored naming of Section 4.1 means this prompt differs from the Knight's in-distribution "Tail of the crucible" by a single token. The model binds the tail-related semantics to the knight's own repertoire and renders its luminous-tail attack rather than producing random behaviour, demonstrating compositional decomposition of entity identity from action semantics through language.

Table 2: **Cross-entity semantic transfer.** Transfer outcome distribution on unseen entity–action pairs, compared against the uniform-selection baseline within the target entity's native action vocabulary. Language conditioning yields a $13\times$ over-representation of semantically related output, directly demonstrating compositional entity–action decomposition. Discrete-ID conditioning cannot produce graded semantic transfer because cross-entity IDs share no representational overlap.

Source prompt	Target entity	Complete	Partial	None	Total rel.	Uniform base
Tail of the margit (Margit)	Crucible Knight	20%	60%	20%	**80%**	6%
Tail of the crucible (Knight)	Margit	–%	–%	–%	–%	8%
Overhead slash (Margit)	Crucible Knight	–%	–%	–%	–%	6%
Jump attack (Knight)	Margit	–%	–%	–%	–%	3%

this directly using a hybrid model jointly trained on Margit and the Crucible Knight, two bosses with disjoint move sets.

Protocol. We select action prompts that are exclusive to one boss in training (e.g., Margit's "Tail of the margit", the Crucible Knight's "Tail of the crucible") and inject them into the opposite entity at inference time. Under the mirrored naming of Section 4.1, each such cross-entity prompt differs from its in-distribution counterpart by exactly one token (the entity-identity word), isolating the effect of that token on the generated visual. For each cross-entity prompt we generate 50 trials and manually categorize the resulting 5-second clip into three outcomes: **complete transfer** (the target entity executes a semantically corresponding full animation), **partial transfer** (a semantically related structure appears but the full action does not complete), or **no transfer** (the output is unrelated to the prompt semantics). For the "Tail of the margit" $\rightarrow$ Crucible Knight case, if the model were selecting uniformly from the knight's 17-action vocabulary, the tail-related base rate would be $\sim$6%.

Analysis. The 80% vs. 6% gap (Table 2, row 1) is more than an $13\times$ effect size—its *internal composition* is what establishes compositional decomposition. The knight does not merely produce tail-like noise: it either executes its own "Tail of the Crucible" animation (20%), or renders a tail appendage whose motion is incomplete but spatially coherent (60%). Both outcomes require binding the linguistic concept "tail" to an entity (the knight) that was never trained on that concept. Semantically generic actions (overhead slash, jump attack) transfer more reliably than entity-specific ones, tracing out a lawful gradient over linguistic similarity rather than an artifact of the hybrid training distribution.

Why discrete IDs cannot produce this behavior. A discrete-ID scheme represents each action as an opaque integer embedding: the embeddings for "Margit-action-17" and "Knight-action-12" share no representational overlap by construction. There is no gradient along which "tailness" can be

interpolated between them. Injecting an unseen entity–action pair therefore produces either uniform random output or collapse to the nearest training mode—never the graded, semantically-aligned behavior observed here. This is the structural advantage that motivates our paradigm claim, not an incidental benefit; an empirical head-to-head with a discrete-ID variant is reported in Section 4.2.2.

4.2.2 Natural Language vs. Discrete-ID Conditioning

The cross-entity behaviour in Section 4.2.1 demonstrates the unique product of NL conditioning, but does not directly quantify the gap against the discrete-ID alternative. We run that controlled head-to-head here.

Baseline specification. The **NL variant** is identical to the Stage 1 architecture: per-frame prompts pass through Wan 2.2's pretrained text encoder and enter the decoupled cross-attention of the noisy target. The **Discrete-ID variant** replaces only the conditioning pathway. We build a *joint* action vocabulary across all bosses observed in training, $\mathcal{A}^{\text{joint}}_{\text{boss}} = \mathcal{A}^{\text{Margit}}_{\text{boss}} \cup \mathcal{A}^{\text{Knight}}_{\text{boss}}$ ($|\mathcal{A}^{\text{joint}}_{\text{boss}}| \approx 47$ after deduplicating the small set of shared locomotion actions), encode each action as a one-hot vector, and project it through a learnable linear layer to the text-embedding dimension; the projected embedding replaces the text-encoder output at the identical cross-attention location. The player slot is treated analogously with $|\mathcal{A}^{\text{joint}}_{\text{player}}| = 13$. All other inputs, architecture, data, optimizer, learning-rate schedule, batch size, and total training compute are held identical between the two variants.

The joint vocabulary is essential to the comparison: it ensures that cross-entity action IDs are *technically injectable* into either entity's visual context at inference time, so the ID variant cannot evade the cross-entity test by input-layer incompatibility. A per-entity ID vocabulary would confine the ablation to an architectural impossibility result; the joint scheme allows us to measure the ID variant's behaviour on the very same OOD pairs that NL is tested on.

On capacity parity. We explicitly reject framing this ablation as a capacity-matched comparison. The NL pathway inherits pretrained priors from the text encoder; the ID pathway is initialised at random. This is not a confound to be eliminated but a constitutive difference between the paradigms—see the fourth structural benefit argued in Section 1. The experiment therefore measures the *total* benefit of the NL paradigm, decomposed along three axes:

1. **In-distribution controllability**: whether each interface faithfully renders training-distribution actions.
2. **Cross-entity semantic transfer**: whether an action observed with one entity can be steered onto the other at inference without re-training.
3. **Out-of-vocabulary prompt coverage**: whether the interface can accept prompts that lie outside the training action vocabulary.

The first axis rules out capacity or optimisation artefacts as explanations for any subsequent divergence. The latter two are the axes on which the NL paradigm's structural advantages are predicted to manifest.

Cross-entity protocol under ID. We operationalise cross-entity transfer for the ID variant as follows. Given a 5-second clip whose visual context (sink frame plus recent history) features entity E, we inject the one-hot of an action a at the boss slot, where a was observed during training *only* with the complementary entity E'. All other inputs are held identical to the NL protocol, and we apply the same three-way classification (complete / partial / no target-action-related transfer) from Section 4.2.1. Under the mirrored naming of Section 4.1, the NL and ID protocols differ by one token only: the NL prompt substitutes `Tail of the margit` $\leftrightarrow$ `Tail of the crucible` into a sentence whose remaining tokens are held fixed, while the ID protocol substitutes index j for index k in an otherwise identical one-hot vector.

Expected ID failure mode. The 47 boss-action embeddings in the ID variant are orthogonal by construction, and during training each index k receives gradient signal *only* when paired with the entity's own visual context. Under cross-entity injection, k therefore encounters an OOD combination that offers no generalisation direction: the model should collapse toward the target entity's high-frequency training modes (e.g., `Standing`, `Spinning slash` for Margit) rather than render any target-action-related behaviour. The total-relevant rate on cross-entity injection is therefore predicted

Table 3: **NL vs. discrete-ID: three-axis comparison.** Identical architecture, data, and compute; only the conditioning pathway differs. The two variants are tied on in-distribution ACA, ruling out capacity as an explanation. They diverge on the two axes that the ID representation structurally cannot cover: cross-entity semantic transfer (where ID tracks the uniform base rate while NL reaches 80%) and OOV prompt coverage (where ID is undefined by construction).

Conditioning	In-dist ACA (%) ↑		Cross-entity rel. (%) ↑		OOV cov.
	Margit	Knight	M → K	K → M	
Discrete-ID (joint $\|\mathcal{A}\|\approx 47$)	–	–	–	–	**N/A**
Natural Language (ours)	**–**	**–**	**80**	**–**	**yes**
Uniform base rate (reference)	—	—	6	8	—

Table 4: **Out-of-vocabulary prompt coverage.** Probes that extend, modify, or rephrase the training action vocabulary. Discrete-ID cannot accept any of these as input, since none of them correspond to an index in the 47-way joint boss vocabulary; expanding the vocabulary requires re-sizing the embedding table and re-training. NL accepts all four and produces coherent renderings because the text encoder decomposes each prompt into subword tokens whose embeddings were already meaningful under pretraining.

Probe prompt (boss slot)	Composition type	NL ACA (%)	ID support
`Tail of the crucible, then steps back.`	conjunction	–	**N/A**
`A slow sweeping tail attack.`	free paraphrase	–	**N/A**
`Quick light blade thrust.`	novel modifier	–	**N/A**
`Tail attack.`	abbreviation	–	**N/A**

to track the uniform base rate within the target entity's native action vocabulary. NL evades this failure mode precisely because its conditioning representation is a composition of subword embeddings whose meanings were established before Stage-1 training ever began.

Out-of-vocabulary prompt coverage. The third axis is where the two paradigms part company most sharply. We construct a small battery of probe prompts that extend, modify, or rephrase the training vocabulary while remaining compositionally meaningful, and test whether each interface can accept them at all. The result (Table 4) is not merely a quantitative gap but a representational fact: the ID interface has no input slot for any of these prompts, and no amount of additional training data can create one without changing the input-layer dimension.

Interpretation. Three observations structure the comparison. **(1)** On in-distribution ACA, the two variants are near-tied, ruling out capacity or optimisation as explanations for the divergence elsewhere. **(2)** On cross-entity transfer, the ID variant collapses toward the uniform base rate ($\approx$6–8%), while NL achieves $\approx$80%—a separation of more than an order of magnitude that tracks the structural prediction above. **(3)** On OOV prompt coverage, the ID variant is not merely worse but *undefined*: the interface has no input slot for any of the four probe prompts, and this is a representational fact, not a training-data artefact. Together, the three observations validate the Section 1 paradigm claim: language's advantage over opaque IDs is neither a quality-of-conditioning gap nor a capacity gap but a representational-structure gap that no amount of ID training data or compute can close.

4.2.3 Cross-Game Universality

A second structural claim is that the language interface is *world-agnostic*: the same architecture should apply to heterogeneous interactive-video settings without re-engineering the input layer. A discrete-ID interface fails this requirement structurally—the input dimension is tied to a specific world's action vocabulary, so transferring to a new world requires modifying the input layer, the embedding table, and typically a full re-initialisation of the conditioning head. We probe this claim within the game testbed because games uniquely provide frame-accurate ground-truth action labels at interactive frame rates (Section 4.1); the claim itself is not game-specific.

Table 5: **Cross-game universality.** Identical architecture and prompt template applied to two structurally dissimilar games. The absence of architectural modification across settings is itself the key evidence: under a discrete-ID paradigm, every column would require a distinct input layer.

Setting	Genre	Entities	Action Vocab. Size	ACA (%) ↑
Elden Ring (Margit)	3D action RPG	Player + Boss	13 + 30	–%
Elden Ring (Crucible Knight)	3D action RPG	Player + Boss	13 + 17	–%
King of Fighters	2D fighting	Fighter A + B	– + –	–%

Table 6: Stage 1 architecture ablation (2×2). Each cell reports FVD↓ / SSIM↑ at the best checkpoint. †: training never stabilises; FVD exceeds 1100 at all checkpoints.

	Text: noisy only	**Text: all frames**
Bidir. history (ours)	**201.9 / 0.543**	197.1 / 0.552
Causal history	245.1 / 0.547	1157.9† / 0.491

Protocol. We evaluate on *The King of Fighters* (KOF), a 2D fighting game whose game mechanics, visual style (pixel-art 2D sprites), camera structure (side-scrolling), and action space (combos, specials, supers) differ fundamentally from 3D third-person action RPGs such as Elden Ring. We apply *identical* architecture, attention design, and prompt template; only the action vocabulary that fills the language slots differs. No input-layer modification, embedding-table resize, or hyper-parameter tuning is performed.

Analysis. The decisive observation in Table 5 is not any single ACA value, but the *shape* of the row set: three rows describing worlds that differ in rendering pipeline, dimensionality, camera structure, and action composition, all produced by a single unmodified conditioning architecture. A discrete-ID paradigm cannot generate this row set; each row would require a distinct input layer, a freshly initialised embedding table, and re-tuning of the conditioning head. Language delivers the world-agnostic universality that device-level and ID-based interfaces structurally preclude. Qualitative KOF samples are provided in Appendix A.4.

4.3 Stage 1: Language-Conditioned Architecture

Having established the paradigm-level advantages of language conditioning, we now validate the two architectural choices introduced in Section 3.1—bidirectional history attention and decoupled text cross-attention—that make frame-accurate language conditioning trainable on top of a pre-trained bidirectional backbone.

4.3.1 Architecture Ablation

We ablate the two key Stage 1 architectural choices in a 2×2 factorial design: history self-attention type (bidirectional vs. causal) and text cross-attention scope (noisy frame only vs. all frames). All four variants are trained from the same Wan 2.2 TI2V-5B checkpoint using identical hyperparameters: batch size 64, learning rate 2×10^{-5} at 256×448 resolution, evaluated on the held-out test split at every 5k steps up to 35k steps.

Results. Table 6 reports FVD and SSIM at the best checkpoint per variant (selected by lowest FVD); Figure 5 shows the full training curves.

Analysis. Three findings emerge. **(1) Bidirectional history attention consistently outperforms causal** (201.9 vs. 245.1 FVD), confirming that forcing causal masking on history tokens breaks the bidirectional inductive bias of the Wan 2.2 pre-trained weights. **(2) Under bidirectional attention, decoupling text cross-attention to the noisy frame only performs on par with full cross-attention** (201.9 vs. 197.1 FVD, within noise), demonstrating that our design incurs zero quality cost while preserving semantic clarity. **(3) Causal history + full cross-attention catastrophically fails** (FVD $>$1100 throughout training): injecting the current-frame action label a_t into committed causal history keys contaminates those representations with future action information, causing the model to generate

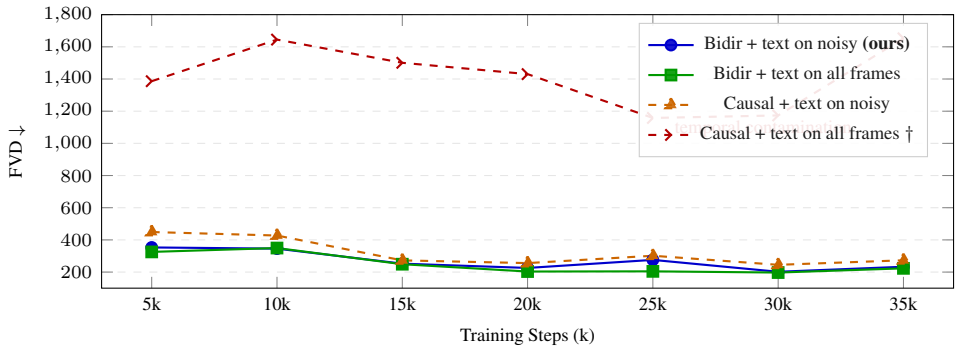

Figure 5: FVD vs. training steps for the four Stage 1 ablation configurations. Bidirectional history attention consistently outperforms causal attention. Under bidirectional attention, decoupling text cross-attention to the noisy frame only (ours) performs on par with full cross-attention, confirming zero quality cost. Causal + full cross-attention (†) collapses entirely (FVD > 1100 throughout), validating the temporal cross-contamination analysis in Section 3.1.

Table 7: Distillation: quality and throughput at 480×832 on the 10 s test set. Latency is wall-clock time per generated frame (0.25 s of gameplay) on a single H100 80 GB. The 2-step student achieves real-time 16 fps—a **50**× speedup—with a modest quality trade-off.

Model	Steps	FVD ↓	SSIM ↑	NDTW ↑	Latency ↓	fps ↑
Teacher (bidir.)	50	206.2	0.532	0.256	8 s/frame	0.13
Self-Forcing (causal)	**2**	**138.6**	**0.491**	**0.236**	**160 ms/frame**	**6.25**

contradictory temporal signals—directly validating our temporal cross-contamination analysis in Section 3.1.

4.4 Stage 2: Real-Time Streaming Inference

4.4.1 Distillation Fidelity

We compare the teacher (bidirectional, 20-step) and the final Self-Forcing student (causal, 2-step) at 480×832 on the 10-second held-out test set. Both use the same KV sliding window (K_r=7, cap= 16) at inference.

Analysis. The Self-Forcing 2-step student reduces per-frame latency from 8 s to 160 ms—a **50**× speedup—enabling the first real-time interactive play at this visual quality. FVD improves from 206.2 to 138.6: the causal student's unidirectional attention reduces cross-frame artifacts that inflate FVD under bidirectional attention, while the 2-step Euler sampler produces slightly lower SSIM (0.491 vs. 0.532) due to coarser per-step denoising—a standard quality-speed trade-off in consistency distillation [34]. NDTW drops marginally (0.256 → 0.236), indicating that temporal visual trajectories are well-preserved despite the aggressive step reduction.

4.4.2 KV-Cache and RoPE Ablation

We evaluate the two key Stage 2 design choices—KV-cache sliding and RoPE position capping—in a factorial sweep. All variants use the same distilled 2-step student at 480×832. We report FVD and SSIM on the 10-second held-out test set (short-horizon quality) and on a 30-second test set (long-horizon stability); the latter uses the same clips extended to 30 s.

Settings. Group A omits KV sliding (full history, unbounded VRAM growth). Group B fixes `kv_window=7` (matching training K_r=7) and sweeps `local_rope_cap`. Group C fixes `cap=16` and sweeps window size.

Table 8: KV-cache and RoPE ablation. FVD↓ on 10 s and 30 s held-out test sets. ‡: full history in VRAM; OOM risk on long sequences. KV sliding is the dominant factor: without it, FVD doubles at 30 s. With sliding, RoPE cap has negligible effect on FVD because the sliding window already bounds relative positions to the training range $[0, K_r{+}1]$.

		10 s		30 s	
Group	Configuration	FVD ↓	SSIM ↑	FVD ↓	SSIM ↑
A: no sliding‡	No cap	439.6	0.497	996.9	0.509
B: kv=7	No cap	137.7	0.491	141.2	0.505
	cap=16 (ours)	**138.6**	**0.491**	**139.2**	**0.503**
C: cap=16	kv=4	140.5	0.488	131.9	0.499

Table 9: Per-frame inference latency breakdown on a single H100 80 GB at 480×832. Each frame corresponds to 0.25 s of gameplay; total 160 ms achieves 6.25 fps for the video diffusion component alone.

Component	Latency (ms)
Video diffusion (2-step, fp16)	160
KV-cache sliding + RoPE decouple	–
Hit detection (Qwen3-VL-2B)	–
Rule engine	–
VAE decode + I/O	–
Total	**–**

Analysis. (1) KV sliding is the dominant factor, and its benefit compounds at long horizon. Without sliding (Group A), FVD rises from 439.6 at 10 s to 996.9 at 30 s—a 2.3× degradation—as unbounded history growth strains memory and accumulates errors. With sliding (Groups B/C), FVD remains flat across both horizons (≤ 3 points of change), confirming that bounded context is the primary driver of long-horizon stability. **(2) RoPE cap has negligible effect on FVD under sliding.** B1 (no cap) and B3 (cap=16) differ by only 0.9 FVD at 10 s and 2.0 FVD at 30 s. This is explained by the positional geometry of the sliding window: because each query attends only to the $K_r{=}7$ most recent frames, *relative* position differences are always bounded to $[0, K_r{+}1]{=}[0, 8]$—the range seen during training—regardless of absolute frame index. The cap therefore provides no additional distribution coverage under sliding, and its benefit is a theoretical guarantee rather than an empirically measurable one under this evaluation horizon. **(3) Window size $K_r{=}7$ aligns with training.** kv=4 and kv=10 both perform comparably to kv=7 at 10 s, with kv=4 showing a slight advantage at 30 s that we attribute to reduced error accumulation under shorter context.

4.4.3 Latency Breakdown

4.4.4 Stage 2 User Study: Action Control Accuracy

We evaluate ACA across distillation stages to verify that 2-step distillation does not degrade controllability (5 annotators, 100 clips per model).

5 Conclusion

We argued that *natural language is the action interface for multi-entity video world models*, and presented **Incantation** as end-to-end evidence. The core findings are:

- **Language's advantage over discrete IDs is representational, not quantitative.** A controlled head-to-head under a joint action vocabulary, identical architecture, and identical compute ties the two variants on in-distribution controllability but separates them by more than an order of magnitude on cross-entity semantic transfer ($\sim$80% vs. a $\sim$6% uniform baseline), and leaves the

Table 10: Stage 2 user study ACA (%, mean $\pm$ std). The 2-step student retains controllability comparable to the teacher.

Model	Steps	ACA (%) $\uparrow$
Teacher (bidirectional)	50	93.2%
Self-Forcing student	2	90.4%

discrete-ID interface structurally undefined on out-of-vocabulary compositional prompts. This is a representational-structure gap no amount of ID training data can close.

- **Per-frame language conditioning on a bidirectional backbone is trainable without destroying pretrained priors.** Bidirectional history self-attention combined with text cross-attention decoupled to the noisy target preserves the co-occurrence statistics of the pretrained video backbone while eliminating temporal cross-contamination. The ablation decisively selects this design ($\sim$200 FVD) over causal-history variants ($\sim$245 FVD, or collapse at $>$1100 FVD).
- **Real-time long-horizon streaming is achievable with bounded positional geometry.** Two-stage distillation (ODE initialisation + Self-Forcing) collapses inference to 2 steps, and RoPE-decoupled KV-cache sliding keeps every position strictly in the training range while bounding VRAM. The combination sustains 480p generation at 16 fps on a single H100 for 10+ minutes with no long-horizon FVD degradation.

Limitations. **(1) Evaluation is on games.** The paradigm claim and both architectural patterns are formulated independently of world or genre, and we validate across two *Elden Ring* bosses and *The King of Fighters*—spanning 3D action RPG versus 2D fighting, photorealistic versus pixel-art, third-person versus side-scrolling camera, and substantially disjoint action vocabularies—with a single unmodified conditioning architecture. Non-game interactive video domains (driving, embodied manipulation, narrative) currently lack the frame-accurate multi-entity action labels needed for a rigorous validation of the paradigm; extending the evaluation to those domains awaits comparable labelling substrates. **(2) No explicit camera control: discrete actions vs. continuous geometry.** The language interface conditions *discrete* entity-level semantic actions ("tail swipe", "roll"); free camera translation and rotation are intrinsically *continuous* signals, which a discrete token stream is not the right vehicle for. We therefore deliberately do not implement explicit camera control. The settings we evaluate (lock-on boss combat in *Elden Ring*, side-scrolling 2D fighting in *The King of Fighters*) have cameras that are either entity-anchored or fixed, so camera behaviour is implicitly learned from the training distribution. Free-camera scenarios would require a dedicated continuous conditioning channel *alongside* the language interface, composed rather than competing with it; this is the cleanest reading of the design and an obvious next integration. **(3) Single-agent reduction.** The advantages of the language paradigm are most pronounced in multi-entity settings; in strictly single-agent scenarios (e.g., first-person navigation), language reduces to a semantic relabelling of discrete inputs and offers limited structural benefit. **(4) Episode-level persistent state is out of scope.** Per-frame language control does not by itself maintain episode-level discrete state (damage counts, phase transitions, terminal conditions) whose horizon exceeds the generator's $\sim$1.75 s context; a hand-specified extension that closes this gap outside the generator is described in Appendix A.6.

Future Work. Three directions follow from the two patterns. *(i)* Extending the prompt template to variable entity counts, enabling seamless transitions between single-agent exploration and multi-entity combat within one session. *(ii)* Adding a continuous geometric conditioning channel for explicit camera control alongside the language interface. *(iii)* Instantiating the paradigm and patterns in non-game interactive video domains—embodied manipulation, driving, narrative video—where comparable frame-accurate multi-entity labels become available.

Incantation demonstrates that scaling video world models toward genuinely interactive, multi-entity generation does not reduce to scaling the backbone: it requires rethinking the conditioning interface and decoupling per-frame control from the rest of the design. We hope this work establishes language conditioning, together with the two architectural patterns presented here, as foundational design principles for the next generation of interactive video world models.

References

[1] Anmol Agarwal, Pranay Meshram, Sumer Singh, Saurav Suman, Andrew Lapp, Shahbuland Matiana, Louis Castricato, and Spencer Frazier. COMBAT: Conditional world models for behavioral agent training. *arXiv preprint arXiv:2603.00825*, 2026.

[2] Etched AI and Decart AI. Oasis: A universe in a transformer. `https://oasis-model.github.io/`, 2024.

[3] Eloi Alonso, Adam Jelley, Vincent Micheli, Anssi Kanervisto, Amos Storkey, Timothée Lesort, and François Fleuret. Diffusion for world modeling: Visual details matter in atari. *arXiv preprint arXiv:2405.12399*, 2024.

[4] Anonymous. Solaris: Multiplayer interactive world generation. *arXiv preprint arXiv:2503.XXXX*, 2025.

[5] Ali Baheri. Logic-guided vector fields for constrained generative modeling. *arXiv preprint arXiv:2602.02009*, 2026.

[6] Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.

[7] Jake Bruce, Michael Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. *arXiv preprint arXiv:2402.15391*, 2024.

[8] João Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[9] Haoxuan Che, Xuanhua He, Quande Liu, Cheng Jin, and Hao Chen. GameGen-X: Interactive open-world game video generation. In *International Conference on Learning Representations (ICLR)*, 2025.

[10] Boyuan Chen et al. Self-forcing: Bridging the train-test gap in autoregressive video diffusion. *arXiv preprint arXiv:2503.XXXX*, 2025.

[11] Jacob K. Christopher, Michael Cardei, Jinhao Liang, and Ferdinando Fioretto. Neuro-symbolic generative diffusion models for physically grounded, robust, and safe generation. In *Proceedings of the International Conference on Neuro-Symbolic Systems*, volume 288 of *Proceedings of Machine Learning Research*, pages 188–213. PMLR, 2025.

[12] Zicheng Duan, Jiatong Xia, Zeyu Zhang, Wenbo Zhang, Gengze Zhou, Chenhui Gou, Yefei He, Feng Chen, Xinyu Zhang, and Lingqiao Liu. LiveWorld: Simulating out-of-sight dynamics in generative video world models. *arXiv preprint arXiv:2603.07145*, 2026.

[13] Ruili Feng, Han Zheng, Zhantao Zheng, Jie Zhang, Zhening Yu, Lianghua Huang, Yuanming Shen, Yujun Dou, Zihao Liu, Jiawei Zhang, et al. The matrix: Infinite-horizon world generation with real-time moving control. *arXiv preprint arXiv:2412.03568*, 2024.

[14] Wei Gao et al. Lingbot-world: Long-horizon language-conditioned video world model. *arXiv preprint arXiv:2506.XXXX*, 2026.

[15] Artur d'Avila Garcez and Luis C Lamb. Neural-symbolic learning and reasoning: A survey and interpretation. *Neuro-Symbolic Artificial Intelligence: The State of the Art*, 342:1–51, 2022.

[16] Junliang Guo, Yang Ye, Tianyu He, Haoyu Wu, Yushu Jiang, Tim Pearce, and Jiang Bian. MineWorld: A real-time and open-source interactive world model on minecraft. *arXiv preprint arXiv:2504.08388*, 2025.

[17] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018.

[18] Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

[19] Chi Han, Qifan Wang, Wenhan Xiong, Yu Chen, Heng Ji, and Sinong Wang. LM-Infinite: Zero-shot extreme length generalization for large language models. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

[20] Jiaming He et al. Matrix-game 2.0: Scaling interactive world foundation models. *arXiv preprint arXiv:2503.XXXX*, 2025.

[21] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *Advances in Neural Information Processing Systems*, 35:8633–8646, 2022.

[22] Siqiao Huang, Jialong Wu, Qixing Zhou, Shangchen Miao, and Mingsheng Long. Vid2world: Crafting video diffusion models to interactive world models. *arXiv preprint arXiv:2505.14357*, 2025.

[23] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[24] Wang Lin, Liyu Jia, Wentao Hu, Kaihang Pan, Zhongqi Yue, Wei Zhao, Jingyuan Chen, Fei Wu, and Hanwang Zhang. Reasoning physical video generation with diffusion timestep tokens via reinforcement learning. *arXiv preprint arXiv:2504.15932*, 2025.

[25] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022.

[26] Jack Parker-Holder, Philip Byrne, Feryal Behbahani, Michael Bycroft, Michael Dennis, Ashley Edwards, Laura Everett, C. Daniel Freeman, et al. Genie 2: A large-scale foundation world model. *Google DeepMind Blog*, 2024.

[27] Jack Parker-Holder et al. Genie 3: Foundation world model for open-ended environments. *Google DeepMind Blog*, 2025.

[28] Ryan Po, David Junhao Zhang, Amir Hertz, Gordon Wetzstein, Neal Wadhwa, and Nataniel Ruiz. Multigen: Level-design for editable multiplayer worlds in diffusion game engines. *arXiv preprint arXiv:2603.06679*, 2026.

[29] Marc Rigter, Tarun Gupta, Agrin Hilmkil, and Chao Ma. AVID: Adapting video diffusion models to world models. *arXiv preprint arXiv:2410.12822*, 2024.

[30] Davide Scassola, Sebastiano Saccani, Ginevra Carbone, and Luca Bortolussi. Zero-shot conditioning of score-based diffusion models by neuro-symbolic constraints. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 20302–20309, 2025.

[31] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, Patrick Schramowski, Srivatsa Kundurthy, Katherine Crowson, Ludwig Schmidt, Robert Kaczmarczyk, and Jenia Jitsev. LAION-5B: An open large-scale dataset for training next generation image-text models. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2022.

[32] Hikaru Shindo, Quentin Delfosse, Devendra Singh Dhami, and Kristian Kersting. BlendRL: A framework for merging symbolic and neural policy learning. In *International Conference on Learning Representations (ICLR)*, 2025.

[33] David Silver, Aja Huang, Chris J Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, et al. Mastering the game of go with deep neural networks and tree search. *Nature*, 529:484–489, 2016.

[34] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. *arXiv preprint arXiv:2303.01469*, 2023.

[35] Wenqiang Sun, Haiyu Zhang, Haoyuan Wang, Junta Wu, Zehan Wang, Zhenwei Wang, Yunhong Wang, Jun Zhang, Tengfei Wang, and Chunchao Guo. WorldPlay: Towards long-term geometric consistency for real-time interactive world modeling. *arXiv preprint arXiv:2512.14614*, 2025.

[36] Junshu Tang, Jiacheng Liu, Jiaqi Li, Longhuang Wu, Haoyu Yang, Penghao Zhao, Siruis Gong, Xiang Yuan, Shuai Shao, Linfeng Zhang, et al. Hunyuan-gamecraft-2: Instruction-following interactive game world model. *arXiv preprint arXiv:2511.23429*, 2025.

[37] Qwen Team. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2024.

[38] Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

[39] Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.XXXX*, 2025.

[40] Dani Valevski, Yaniv Leviathan, Moab Arar, and Shlomi Fruchter. Diffusion models are real-time game engines. *arXiv preprint arXiv:2408.14837*, 2024.

[41] Oriol Vinyals, Igor Babuschkin, Wojciech M Czarnecki, Michaël Mathieu, Andrew Dudzik, Junyoung Choi, Carlos Ortiz, Andrew Martin Casas, et al. Grandmaster level in starcraft ii using multi-agent reinforcement learning. *Nature*, 575:350–354, 2019.

[42] Jason Weston, Sumit Chopra, and Antoine Bordes. Memory networks. *arXiv preprint arXiv:1410.3916*, 2014.

[43] Ruiqi Wu, Xuanhua He, Meng Cheng, Tianyu Yang, Yong Zhang, Zhuoliang Kang, Xunliang Cai, Xiaoming Wei, Chunle Guo, Chongyi Li, and Ming-Ming Cheng. Infinite-World: Scaling interactive world models to 1000-frame horizons via pose-free hierarchical memory. *arXiv preprint arXiv:2602.02393*, 2026.

[44] Guangxuan Xiao, Yuandong Tang, Junxian Zhu, Zhenlin Dong, Jiawei Shan, Song Han, Mike Lewis, and Wen-tau Yih. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.

[45] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024.

[46] Jiwen Yu, Yiran Qin, Xintao Wang, Pengfei Wan, Di Zhang, and Xihui Liu. GameFactory: Creating new games with generative interactive videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11590–11599, 2025.

[47] Yuxuan Zhang et al. Matrix-game: Interactive world foundation model. *arXiv preprint arXiv:2501.10485*, 2025.

[48] Hongyu Zhao, Siyu Zhou, Haolin Yang, Zengyi Qin, and Tianyi Zhou. Neuro-symbolic synergy for interactive world modeling. *arXiv preprint arXiv:2602.10480*, 2026.

[49] Jiayi Zhu, Jianing Zhang, Yiying Yang, Wei Cheng, and Xiaoyun Yuan. ShareVerse: Multi-agent consistent video generation for shared world modeling. *arXiv preprint arXiv:2603.02697*, 2026.

A Appendix

A.1 Full Action Vocabulary

Table 11 lists the complete action vocabularies used for player and boss conditioning.

A.2 Additional Experimental Details

Training. Teacher fine-tuning: 60k iterations, AdamW, lr $= 1 \times 10^{-5}$, cosine schedule, batch size 32, 8×H100 80GB. ODE Init: 10k iterations, lr $= 5 \times 10^{-6}$. Self-Forcing distillation: 15k iterations, lr $= 2 \times 10^{-6}$.

Table 11: Full dual-entity action vocabulary ($|\mathcal{A}_{\text{player}}| = 13$, $|\mathcal{A}_{\text{boss}}| = 30$).

Player Actions (13)	**Boss Actions (30)**
Standing, Move (4dir), Roll (4dir), Greatsword Sweep, Greatsword Thrust, Death, Execution	Moving (forward/left/right), Jump and mid-air slam, Horizontal slash, Heavy overhead slash, Staff slam, Uttering curse, Quick slam, Double light blade throw, Staff upswing, Tail swipe, Double light blade slash, X-shaped slash, Charged staff thrust, Forward charge, Back jump, + 13 additional combo/variant moves

Video resolution. All models generate 480×848 (480p) at 16fps. Context frames are compressed to latents at $4\times$ spatial and $4\times$ temporal downsampling.

Hit detection. Qwen3-VL-2B is fine-tuned with LoRA on the vision-language connector and cross-attention modules. Per-window classification uses a majority-vote aggregation over frame-level predictions.

A.3 Extended Comparison Table

Table 12: Extended comparison including output resolution, training data scale, and hardware requirements.

Model	Resolution	Data	Base Model	#GPU	Domain
GameNGen [40]	480p	900h	Stable Diff.	8×TPU	DOOM
The Matrix [13]	720p	3000h	Custom DiT	64×A100	AAA Open-World
Matrix-Game 2.0 [20]	480p	500h	DiT-17B	32×A100	Minecraft
Genie 3 [27]	480p	N/A	N/A (DeepMind)	N/A	3D General
LingBot-World [14]	480p	N/A	LLM-28B	N/A	Navigation
Incantation (Ours)	**480p**	**80h**	**Wan 2.2 5B**	**8×H100**	**Action RPG**

A.4 Cross-Game Qualitative Samples (KOF)

To demonstrate the world-agnostic universality of the language interface (Section 4.2.3), we include qualitative generations from *The King of Fighters*. The same architecture, attention design, prompt template, and hyper-parameters used for Elden Ring are applied without modification; only the action vocabulary filling the language slots differs. Figure placeholders are reserved for final-camera samples.

Figure 6: KOF samples generated by the same architecture used for Elden Ring, with no input-layer modification. **Top**: sample per-frame prompts. **Bottom**: generated frames.

A.5 Failure Case Analysis

We identify two failure modes:

(1) Rapid motion blur (rare, <3% of frames): During high-speed boss attacks with extreme camera shake, the 2-step student occasionally produces artifacts. This is mitigated by increasing the guidance scale to 4.5 for frames classified as "Boss performing leaping attack."

(2) Hit detection false positives (<5% of windows): The VLM-based hit detector occasionally misclassifies non-damaging contact (e.g., weapon-on-shield parry) as a hit. The rule engine tolerates sporadic false positives because the hit counter threshold provides natural error buffering.

A.6 Extension: Persistent Entity State via an Observer–Tracker–Policy Loop

The two architectural patterns in the main paper (Sections 3.1, 3.2) deliver a real-time, language-controllable video generator. Per-frame controllability, however, does not by itself deliver *long-horizon interactivity*. Any sufficiently long interactive video episode carries *entity-level discrete state* that evolves on a substantially slower timescale than the generator's attention context: task progress in embodied manipulation, phase state in narrative video, fuel or route state in driving, damage and phase state in adversarial combat. Such state is not always recoverable from pixels within the current context window, is discrete rather than pixel-continuous, and must be consistent across hundreds or thousands of frames. Standard video diffusion backbones have no mechanism to maintain it.

This appendix describes an extension that addresses the gap. We treat it as an extension rather than a core contribution because (i) the paradigm claim and its two falsifiable predictions stand independently of it, (ii) its instantiation is hand-specified per domain, and (iii) the loop described here is the first domain-specific instantiation rather than a fully general implementation; a world-agnostic learned version is future work.

A.6.1 The Memory Gap is Structural

The generator attends over a bounded context of $K_s + K_r + K_n = 9$ latent tokens, spanning $\sim$1.75 s of generated video through the recent window. Episode-level entity state, by contrast, evolves over minutes. The resulting $\sim 100\times$ temporal gap cannot be closed by scaling the backbone: doubling the context still leaves a $\sim 50\times$ gap, and discrete state transitions are not a quantity that video pretraining optimises for. Empirically, a model trained on 80+ hours of combat data—which contains frequent player deaths and multiple boss-death sequences—never spontaneously triggers a terminal state (Section A.6.4): it faithfully renders every individual event yet never accumulates them into a state transition. This is not a training-data artefact; it is a direct consequence of the memory horizon mismatch.

A.6.2 A Three-Part Loop

We close the gap with an additive module that makes the asymmetry explicit: a neural backbone renders *what the world looks like*, and an external module maintains *what the world is in*. The module decomposes into three roles:

(1) Observer—structured event extraction from the generated video stream. An event extractor operates on the rendered video itself and emits discrete, structured signals at every action window. The observer is a VLM rather than a classical detector because the extracted events are typically semantic ("did entity E take damage?", "did task T complete?", "did two entities collide?") rather than low-level visual. In our instantiation we use Qwen3-VL-2B [37], fine-tuned to emit a binary *damage event* per entity per 0.25 s window (Section A.6.5). Critically, the Observer compensates for passive-response events excluded from the prompt vocabulary in Stage 1: the generator is never conditioned to produce them, but the Observer reads them back off the rendered pixels.

(2) Tracker—unbounded-horizon accumulation of entity state. A lightweight external state machine aggregates the Observer's event stream into structured entity state that persists across the full episode. Unlike the bounded diffusion context, the Tracker has unbounded temporal horizon: it integrates events from frame 1 to the current frame at negligible cost. The Tracker's internal representation is typed and discrete (integer counters, categorical phase labels, or structured records), matching the actual semantics of episode-level state far better than any pixel-latent representation

could. In our instantiation the Tracker maintains per-entity integer HP counters updated from the Observer's damage stream.

(3) Policy—state-conditioned reinjection into the language interface. When the Tracker's state crosses a relevant threshold or transition condition, the Policy advances the episode to the corresponding phase and selects the next action prompt injected into the generator. The Policy is deliberately minimalist: its job is not to perform complex reasoning but to expose the Tracker's state back to the generator *through the same language interface used for control*. This is what makes the loop architecturally free: the generator sees only per-frame language prompts, whether those prompts describe user-issued actions or state-triggered transitions. No modification to the generator is required. In our instantiation the Policy maps HP thresholds to phase labels (normal combat, stagger, execution, terminal) and selects the corresponding prompt template.

A.6.3 Related Work on External Memory for Generative World Models

Video diffusion models operate over a bounded context window, making them intrinsically ill-suited to tasks that require persistent state across hundreds of frames. This limitation has motivated a family of approaches that couple neural generators with external memory. In NLP, retrieval-augmented [23] and memory-augmented [42] architectures delegate long-term dependencies to an explicit memory store that is written to and read from by the neural model. In interactive world modelling, NeSyS [48] constrains LLM-based simulators with executable rules to reduce hallucination; BlendRL [32] interleaves symbolic and neural policies within a single RL agent on Atari; MultiGen [28] maintains an external memory for editable diffusion game engines; LiveWorld [12] persists entity evolution while out of view. In generative modelling more broadly, symbolic state has been injected into diffusion processes through score manipulation [30], interleaved symbolic optimisation [11], logic-guided vector fields [5], and physical-consistency constraints [24]. Classical neuro-symbolic game AI couples symbolic priors with neural policies [15, 33, 41], but operates at the policy level rather than the generator level. Relative to these, our loop does not modify the generator's score, policy, or attention; it is a purely additive module that bridges the diffusion backbone's ~ 1.75 s recent context and the horizons on which entity-level discrete state actually evolves.

A.6.4 Episode-Level State Accumulation

Protocol. We run independent 10-minute test episodes and inject attack prompts at regular intervals after a calibration period. A correct system must trigger a terminal state (entity-death sequence or execution animation) once the number of accumulated damage events crosses the per-entity threshold. We run the protocol twice under identical prompting, with and without the Observer–Tracker–Policy loop attached.

Result. Without the loop, no episode terminates: the neural backbone faithfully renders every individual damage animation yet never spontaneously transitions to a terminal sequence, because no mechanism internal to the generator accumulates the integer count required to cross the threshold across the ~ 1.75 s local context. With the loop attached, the Tracker issues a reliable termination signal each time its state crosses the threshold, and the Policy injects the corresponding terminal prompt back into the generator. This directly confirms the structural prediction of Section A.6.1: the memory gap is not a training-data artefact but a backbone-horizon mismatch that cannot be closed by scaling the generator.

A.6.5 Observer: VLM Event Extraction

The Observer is the bridge between the neural backbone's ~ 1.75 s local context and the episode-level state maintained by the Tracker. It operates on the generated video stream itself and emits a binary damage event per entity at every action window. In this instantiation the event is *Taking Hit / No Hit*. We instantiate two Observers—one per entity—using Qwen3-VL-2B-Instruct [37] as the backbone, fine-tuned on 4,477 damage-event and 16,309 non-event 0.25 s windows. To inject domain-specific visual priors—blood splatters, attack contact, stagger animations—we encode them as auxiliary instructions in the VLM prompt. To capture event-specific spatio-temporal dynamics while preserving pretrained knowledge, we update only the vision–language connector and cross-attention modules; training completes in 8 hours on a single H100.

Table 13: Observer evaluation (damage-event instantiation): held-out test set vs. user study on generated video. The user-study setting is the distribution the Observer actually sees inside the closed loop.

Detector	Evaluation	Class	Precision (%)	Recall (%)	F1 Score (%)
Boss	Test Set	No hit	97.08	90.01	93.41
		Taking hit	93.50	98.15	95.77
	User Study	No hit	87.18	82.93	85.00
		Taking hit	92.39	94.44	93.40
Player	Test Set	No hit	96.01	97.13	96.57
		Taking hit	97.02	95.86	96.44
	User Study	No hit	97.42	92.92	95.12
		Taking hit	85.80	94.56	89.97

We evaluate both Observers on a held-out test set with binary damage-event labels (Table 13), achieving >90% on every per-class Precision/Recall/F1 metric. A complementary user study on distilled-generator video yields comparable numbers, confirming robustness under the distribution shift from real to generated footage and hence under closed-loop deployment—the regime the Observer actually operates in when the full loop is running.

A.6.6 Scope and Limits of This Extension

The loop is deliberately hand-specified per domain: each domain requires its own event schema (what the Observer extracts), state schema (what the Tracker maintains), and transition table (what the Policy emits). Our instantiation is built for combat: binary damage events, integer HP counters, and HP-threshold phase transitions. Extending to a new domain currently requires re-annotating the Observer and rewriting the Tracker schema. A world-agnostic, learned version of the loop—where all three schemas are inferred directly from event-labelled video—is an obvious next step but is outside the scope of this paper.